

# Oral Exam

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# Projects

I will briefly discuss the three following projects:

- ▶ Invariant measures for the Stochastic Heat Equation with rough initial conditions,
- ▶ Exact moment asymptotics and Solvability for the Interpolated Stochastic Heat and Wave Equation,
- ▶ Global solutions for the Interpolated Stochastic Heat and Wave Equation with a super linear diffusion term and no drift.

Project 1  
Invariant Measures for the Stochastic Heat Equation

# Invariant Measures for the Stochastic Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = b(x, u(t, x)) \dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

We assume that:

- ▶  $|b(x, u) - b(x, v)| < L_b |u - v|$  and  $|b(x, 0)| \leq L_0$  for  $u, v \in \mathbb{R}$  and  $x \in \mathbb{R}^d$ ,
- ▶  $W$  is a centered Gaussian noise that is homogeneously colored in space,

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where  $\tilde{\psi}(s, x) = \psi(s, -x)$  and  $\Gamma$  is a non-negative and nonnegative definite measure referred to as the *spatial correlation measure*.

## Mild Solution

We interpret the solution of the Stochastic Heat Equation as the following stochastic integral equation:

$$\begin{aligned}u(t, x) &= \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y) b(u(s, y)) W(ds, dy) \\&= J(t, x) + I(t, x),\end{aligned}$$

where  $G(t, x)$  is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

A processes,  $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ , that satisfies this relation is referred to as the *mild solution*.

## Existence and Uniqueness

Existence and uniqueness of a mild solution of the Stochastic Heat Equation has been proven under the following two conditions:

- ▶ the initial measure,  $\mu$ , satisfies

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \mu(dx) < \infty \quad \text{for any } a > 0,$$

- ▶ the Fourier transform of the spatial correlation measure,  $\hat{f} := \mathcal{F}\Gamma$  satisfies *Dalang's condition*:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

We refer to  $\hat{f}$  as the spectral measure of the noise  $\dot{W}$ .

## Admissible Weight

Since we work in the whole space  $\mathbb{R}^d$ , we consider weights,  $\rho$  such that

- ▶  $\rho \in L^1(\mathbb{R}^d)$  is positive, continuous and bounded,
- ▶ for any  $T > 0$ , there exists a  $C(T) > 0$  such that

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho)(x)}{\rho(x)} \leq C(T).$$

We then work in the weighted Hilbert Space,  $L^2_\rho(\mathbb{R}^d)$ , with norm defined as

$$\|\phi(\cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} \phi(x)^2 \rho(x) dx.$$

Some examples of admissible weights include the following:

$$\frac{1}{1 + |x|^a}, \quad a > d \quad \text{and} \quad \exp(-a|x|), \quad a > 0.$$

# Intermittency

We define the upper and lower Lyapunov exponents

$$\underline{m}_2 := \liminf_{t>0} \frac{1}{t} \log \left( \mathbb{E}(|u(t, x)|^2) \right), \text{ and } \overline{m}_2 := \limsup_{t>0} \frac{1}{t} \log \left( \mathbb{E}(|u(t, x)|^2) \right).$$

The solution is said to be *fully intermittent* when

$$\inf_{x \in \mathbb{R}^d} \underline{m}_2 > 0.$$

We say a *phase transition* happens when there exists  $\underline{\lambda} \leq \overline{\lambda}$  such that

$$\begin{cases} \sup_{x \in \mathbb{R}^d} \overline{m}_2 = 0 & L_b < \underline{\lambda} \\ \inf_{x \in \mathbb{R}^d} \underline{m}_2 = 0 & \overline{\lambda} < l_b \end{cases}$$

## Bounded Moments

Under the assumptions for existence and uniqueness, we have the following moment bound

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)]H(t; \gamma_p)^{1/2}, \quad \gamma_p = 32pL_b^2,$$

where  $t \mapsto H(t, \gamma_p)$  is nondecreasing. Following a strategy laid out by Daprato and Zabczyk, we wish to prove that

$$\sup_{t \geq 0} \mathbb{E} \left( \|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Hence, a phase transition must occur. By Theorem 1.3, this forces us to require  $d \geq 3$  and  $\Upsilon(0) < \infty$ . If we further assume that  $L_b$  is small then we may also say

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)].$$

## Invariant Measures

Consider a measure space  $(E, \mathcal{E})$  and denote  $\mathcal{M}_1(E)$  to be the set of probability measure on this space.

### Definition (Invariant Measure)

A measure  $\eta(\cdot) \in \mathcal{M}_1(E)$  is invariant for transition functions  $\{P_t(x, \Gamma)\}_{t \geq 0}$  if for any  $\Gamma \in \mathcal{E}$  we have that

$$\eta(\Gamma) = \int_E P_t(x, \Gamma) \eta(dx) \quad \text{for any } t \geq 0$$

Recall from Calculus that the average of a function  $f : [a, b] \rightarrow \mathbb{R}$  is defined as

$$\text{AVG}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$

Thus,  $\eta(\Gamma)$  is the average the family of probability measures  $\{P_t(x, \Gamma)\}_{x \in E}$  for any  $t \geq 0$ .

## Theorem: Existence of an Invariant measure

A processes  $u(t, \cdot)(\omega)$  is bounded in probability in  $L^2_\rho(\mathbb{R}^d)$  if

$$\forall \epsilon > 0, \exists R > 0 : \forall t > 0, \mathbb{P} \left\{ \|u(t, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} \geq R \right\} \leq \epsilon.$$

Tessitore and Zabczyk proved the following:

### Theorem

*Suppose  $d \geq 3$  and that there exists a  $p \in [1, d/(d+2))$  such that the spatial correlation density  $\gamma \in L^p(\mathbb{R}^d)$ . Suppose that there exists two weights,  $\rho$  and  $\hat{\rho}$  and an element  $\phi \in L^2_\rho(\mathbb{R}^d) \cap L^2_{\hat{\rho}}(\mathbb{R}^d)$  such that*

$$\frac{\rho}{\hat{\rho}} \in L^1(\mathbb{R}^d) \quad \text{and} \quad u^\phi(t, \cdot) \text{ is bounded in probability in } L^2_{\hat{\rho}}.$$

*Then there exists an invariant measure in  $L^2_\rho(\mathbb{R}^d)$ .*

## Sketch of the Proof of Existence

It has been shown by DaPrato and Zabczyk that  $P_t(\phi, \Gamma) := \mathcal{L}(u^\phi(t, \cdot))(\Gamma)$  defines a Markov transition function on  $(L_\rho^2(\mathbb{R}^d), \mathcal{L})$  where,

$$\mathcal{L}(u^\phi(t, \cdot))(\Gamma) = \mathbb{P} \left[ \omega \in \Omega : u^\phi(t, \cdot)(\omega) \in \Gamma \right], \quad \Gamma \in \mathcal{L}.$$

We then prove for any  $\epsilon \in (0, 1)$  and  $R > 0$ , the existence of a compact set,  $\mathcal{K} \in \mathcal{L}$  such that

$$\mathcal{L}(u^\phi(t, \cdot))(\mathcal{K}) \geq (1 - \epsilon) \mathbb{P} \left[ \omega \in \Omega : \left\| u^\phi(t - 1, \cdot)(\omega) \right\|_{L_\rho^2(\mathbb{R}^d)} < R \right], \quad t \geq 1.$$

If we assume  $u^\phi(t, \cdot)$  is bounded in probability in  $L_\rho^2(\mathbb{R}^d)$  then one can argue the existence of a sequence  $\{T_n\}$  such that  $0 < T_n \uparrow \infty$

$$\frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u^\phi(t, \cdot))(\Gamma) \xrightarrow[\text{weakly}]{} \eta(\Gamma) \quad \forall \Gamma \in \mathcal{L},$$

and moreover,  $\eta$  is invariant in  $L_\rho^2(\mathbb{R}^d)$ .

## Tessitore and Zabczyk's Further Assumption

If  $B := \sup_{t \geq 0} \mathbb{E} \left( \|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty$  then by Chebyshev's inequality

$$\mathbb{P}[|u(t, \cdot)|_\rho \geq R] \leq R^{-2} \mathbb{E}|u(t, \cdot)|_\rho^2 \leq R^{-2} B < \epsilon, \quad \text{for any } R \geq \sqrt{B/\epsilon}.$$

### Theorem

Suppose that  $d \geq 3$  and that  $L < L_b^{-2}$  where

$$\tilde{\Gamma} := \left| \widehat{\gamma^{1/2}} \right| * \left| \widehat{\gamma^{1/2}} \right| \quad \text{and} \quad L := \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(\zeta) |\zeta|^{2-d} d\zeta < \infty.$$

Then for any admissible weight  $\rho$ , we have that

$$\sup_{t \geq 0} \mathbb{E} \left( \left\| u^1(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

# Moment Bound: Measure Valued Initial Conditions

## Theorem

Suppose that the initial measure,  $\mu$ , and spectral measure,  $\hat{f}$ :

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < \infty \quad \forall t > 0 \quad \text{and} \quad \Upsilon(0) < \infty. \quad (0.1)$$

Then if  $\mu(dx) = \phi(x)dx$  with  $\phi \in L^2_\rho(\mathbb{R}^d)$ ,

$$\sup_{t \geq 0} \mathbb{E} \left( \left\| u^\phi(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Further, if  $\mu$  has no density but we strengthen (0.1) to include

$$\int_0^s dr \, r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty$$

then for any  $t_0 > 0$ ,

$$\sup_{t \geq t_0} \mathbb{E} \left( \left\| u^\mu(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

## Invariant Measures: Measure Valued Initial Conditions

We wish prove the existence of an invariant measure for measure valued initial conditions such as the Dirac delta measure,  $\delta_0(\cdot)$ . Our plan is to consider

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

and then restart our system. In other words, we will consider

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta_x v(t, x) = b(x, v(t, x))\dot{W}(t, x) \\ v(0, \cdot) = u(t_0, \cdot) \end{cases}.$$

With the absence of noise, we know that  $v(t, x) = u(t + t_0, x)$  and we wish to show that this property holds with the addition of noise. We also know that  $\sup_{t \geq 0} \|u(t + t_0, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} < \infty$ . Thus we need to figure out how extend the known proof to handle random initial conditions.

Project 2:  
The Interpolated Stochastic Heat and Wave Equation: Solvability and Exact  
Moment Asymptotics

# The Stochastic Heat and Wave Equations

$$\begin{cases} (\partial_t^b - \Delta) u(t, x) = u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b = 1 \quad (\text{SHE}) \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b = 2 \quad (\text{SWE}) \end{cases}$$

- ▶  $W = \{W(\phi) : \phi \in \mathcal{D}(\mathbb{R}^d)\}$  is a centered and time-independent Gaussian noise
- ▶ The choice of initial condition is such that the solution to the homogeneous equation is constant one.
- ▶ The solution is understood in the *Skorohod* sense.

## Known Results

For the SHE, i.e.,  $b = 1$  ( $a = 2, \nu = 2$ )  
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left( \frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

For the SWE, i.e.,  $b = 2$  ( $a = 2, \nu = 2$ )  
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left( \frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left( \frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

# The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[ \sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (\text{ISHWE})$$

- ▶  $\partial_t^b$  is the *Caputo* fractional derivative
- ▶  $(-\Delta)^{a/2}$  is the fractional Laplacian of order  $a \in (0, 2]$
- ▶  $I_t^r$  is the Riemann-Liouville fractional integral of order  $r \geq 0$

## The Noise, $\dot{W}$ .

The time independent noise informally satisfies

$$\mathbb{E}(\dot{W}(x)\dot{W}(y)) = \gamma(x - y).$$

The spatial correlation and spectral density,  $\gamma$  and  $\varphi$ , for the noise  $\dot{W}$  can be assumed to be any of the following:

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(\xi) = C_1 |\xi|^{d-\alpha} \quad \alpha \in (0, d)$$

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = C_2 \prod_{i=1}^d |\xi_i|^{1-\alpha_i} \quad \alpha_i \in (0, 1), \quad \alpha = \sum_i \alpha_i$$

# Mild Solution

## Definition

For  $T > 0$ , a random field  $u = \{u(t, x) : t \in (0, T), x \in \mathbb{R}^d\}$  is called a *mild solution* if  $G(t - s, x - \cdot)u(s, \cdot)1_{\{s < t\}}$  is Skorohod integrable and the following holds almost surely:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left( \int_{\mathbb{R}^d} G(t - s, x - y) u(s, y) W(\delta y) \right) ds$$

where  $G$  is defined through the *Fox-H function*.

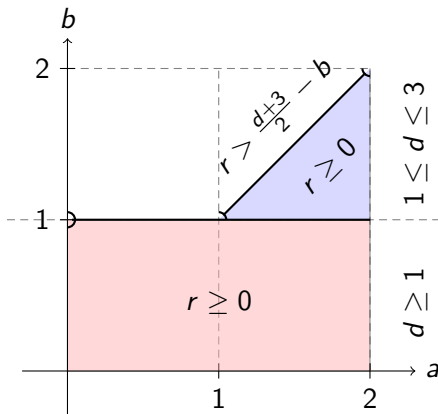
An important characteristic of  $G$  is that

$$\mathcal{FL}G(s, \xi) = \frac{s^{-r}}{s^b + \frac{\nu}{2} |\xi|^a}$$

## Nonnegativity assumption on $G$

Under any of the following cases,  $G$  is nonnegative [Chen, Hu, Nualart '19]:

- ▶  $d \geq 1$ ,  $b \in (0, 1]$ ,  $a \in (0, 2]$ ,  $r \geq 0$ ;
- ▶  $1 \leq d \leq 3$ ,  $1 < b < a \leq 2$ ,  $r > 0$ ;
- ▶  $1 \leq d \leq 3$ ,  $1 < b = a < 2$ ,  $r > \frac{d+3}{2} - b$ .



## Global Solution

$$(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[ \sqrt{\theta} u(t, x) \dot{W}(x) \right]$$

### Definition

$u(t, x)$  is a *global solution* to the (ISHWE) if  $\|u(t, x)\|_p < \infty$  for any  $t > 0$  and  $x \in \mathbb{R}^d$ .

### Theorem (Chen-E. '21+)

*A global solution exists provided  $G$  is nonnegative and*

$$0 < \alpha < \min \left( \frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

# Local Solution

## Definition

$u(t, x)$  is a *local solution* to the (ISHWE) if there exists  $0 < T_a \leq T_b < \infty$  such that  $\|u(t, x)\|_2 < \infty$  when  $0 < t < T_a$  and  $\|u(t, x)\|_2$  D.N.E. (does not exist) for  $t > T_b$ .

## Theorem (Chen-E. '21+)

*A local solution exists provided  $G$  is nonnegative and if*

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r) - 1] \leq d.$$

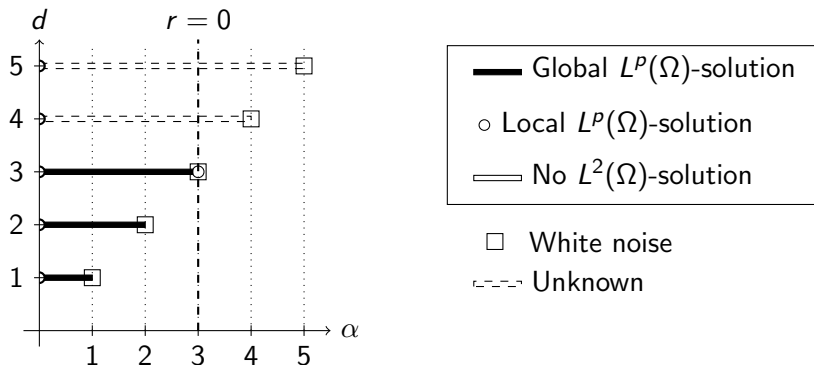
*In this case, a unique  $L^p(\Omega)$  solution exists for  $t \in (0, T_p)$  where*

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}}$$

*and the solution does not exist for  $t > T_2$ .*

## Example: Solvability for the SWE ( $a = b = 2$ )

A local solution only exists when  $\alpha = 3 + 2r \leq d \leq 3$ .

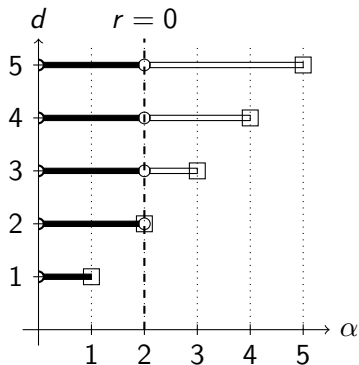


By replacing  $\nu = 2$  in the following, we recover (1.12) [Balan, L. Chen & X. Chen '21]

$$T_p = \frac{\nu^{3/2}}{2\theta(p-1)\sqrt{\mathcal{M}_{2,3}(\delta_0)}}, \quad p \geq 2$$

## Example: Solvability for the SHE ( $a = 2$ and $b = 1$ )

By setting  $a = 2$ ,  $b = 1$  and  $r = 0$ , we obtain the following condition for existence of a local solution:  $\alpha = 2 \leq d$ .



- ▶ When  $\alpha = d = 2$ , the critical time becomes 
$$T_p = \frac{\nu}{2\theta(p-1)\mathcal{M}_{2,2}(\delta_0)}.$$
- ▶ Theorem 4.1 [Y. Hu, '01] proves that an  $L^2(\Omega)$  solution exists for  $t < 2$  but not for  $t > 2\pi$ .
- ▶  $T_2$  being precise implies 
$$2 \leq T_2 = \frac{1}{2\mathcal{M}_{2,2}(\delta_0)} \leq 2\pi$$

# Wiener Chaos Expansion

Through a standard procedure, we may define

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

where  $t = t_{n+1}$  and  $x = x_{n+1}$ , and say that

1.  $u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$
2.  $\mathbb{E}(u(t, x)^2) = \sum_{k=0}^{\infty} \theta^k \|f_k(\cdot, x, t)\|_{\mathcal{H}^{\otimes n}}^2, \quad (t, x) \in (0, T) \times \mathbb{R}^d.$

Through a change of variable, one can easily show that

1.  $\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi)$
2.  $\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$

## How to Find the Critical $\alpha$

For simplicity we consider the SWE ( $a = b = 2$  and  $r = 0$ ).

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \overbrace{t^{[4-\alpha]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2}^{\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2} \\ &\leq \sum_{n \geq 0} \frac{(\theta t^{4-\alpha} c_1 2^{4-\alpha} C_\mu)^n}{(n!)^{3-\alpha}}, \quad C_\mu = \int_{\mathbb{R}^d} \left( \frac{1}{1 + |\xi|^2} \right)^2 \mu(d\xi) < \infty\end{aligned}$$

When  $\alpha = 3$ , then the above reduces down to, we lose the  $n!$  term in the denominator.

## How to Find the Blowup Time (SWE)

Recall from above that the critical case corresponds to  $\alpha = 3 = d$ .

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\ &=: \sum_{n \geq 0} (\theta t)^n R_n, \quad R_n = n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2.\end{aligned}$$

The lemma below says that  $\frac{1}{n} \log R_n \rightarrow \log(2\rho)$ . Now the Cauchy-Hadamard theorem can be directly applied to see that the radius of convergence is precisely  $(2\theta\rho)^{-1} = T_2$ .

Recall the scaling property

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Using this scaling property we see that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Lemma (R. Balan, L. Chen & X. Chen '21)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( \Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left( (n!)^{4-\alpha} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[ \left( \frac{2}{4-\alpha} \right)^{4-\alpha} \rho \right]$$

$\rho$

We define

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[ \int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

Theorem (X. Chen '07)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[ \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left( \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ = \log(\rho_{\nu,a}(\gamma)).$$

where  $\gamma$  is the spatial correlation function.

## Connection of the SPDE with $\rho$

Recall from above that

$$\mathcal{FLG}(1, \xi) = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}.$$

By using this relation, it can be shown that

$$\mathcal{FL}(\tilde{f}_n)(1, \xi) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a}.$$

Applying this to the limit representation of  $\rho$  above yields that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[ \int_{(\mathbb{R}^d)^n} \left| \mathcal{FL}(\tilde{f}_n)(1, \xi) \right|^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu, a}(\gamma)).$$

$\mathcal{M}$ 

We define

$$\begin{aligned}\mathcal{M}_{a,d}(\gamma, \theta) &:= \sup_{g \in \mathcal{F}_a} \left\{ \left( \iint_{\mathbb{R}^{2d}} g^2(x) g^2(y) \gamma(x+y) dx dy \right)^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \gamma \rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\},\end{aligned}$$

where

$$\begin{aligned}\mathcal{E}_a(g, g) &:= (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \quad \text{and} \\ \mathcal{F}_a &:= \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(f, f) < \infty \right\}.\end{aligned}$$

Theorem (Bass, X. Chen and Rosen '09)

$$\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma) < \infty$$

# The Moment Asymptotic

## Theorem (Chen-E. '21+)

*Suppose a global solution to the SPDE in question exists and  $\dot{W}$  is given through the generalized Riesz kernel defined above. Then,*

$$\begin{aligned} & \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ &= \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \end{aligned}$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p := (p-1)^{1-1/\beta} t.$$

# Exact Moment Lyapunov Exponent

(Recall  $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$ )

Corollary (Chen-E. '21+)

For  $p \geq 2$  fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that  $t \mapsto \mathbb{E}(|u(t, x)|^p)$  will grow like  $\exp(K_1 t^\beta)$ , in other words, the moments will grow exponentially in time, and we obtain the exact expression for the constant  $K_1$ .

# Exact Large Moment Asymptotics

(Recall  $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$ )

Corollary (Chen-E. '21+)

For  $t > 0$  fixed, we have that

$$\begin{aligned} & \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= t^\beta \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that  $p \mapsto \mathbb{E}(|u(t, x)|^p)$  will grow like  $\exp(K_2 p^\beta)$ , in other words, the moments will grow exponentially in  $p$  as well.

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## Project 3: Global Solutions for the Interpolated Stochastic Heat and Wave Equation with a Superlinear Drift Term

# The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sigma(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (0.2)$$

The drift coefficient has super-linear growth:

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta \quad \text{as } |x| \rightarrow \infty.$$

We first consider the noise to be space-time white. In other words, the covariance structure is defined for  $\phi, \psi \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$  as

$$J(\phi, \psi) := \mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^d} dx \phi(s, x)\psi(s, x).$$

## Mild Solution

For any  $t \in [0, T]$  and  $x \in \mathbb{R}^d$ , we write

$$u(t, x) = 1 + I(t, x)$$

where

$$I(t, x) = \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy).$$

The function  $Y$  is defined through the Fox-H function and its Fourier transform can be shown to be given as

$$\mathcal{F}Y(t, \cdot)(\xi) = t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha),$$

where

$$E_{\alpha, \beta}(z) = \sum_{k \geq 0} \frac{z^k}{\Gamma(\alpha k + \beta)}.$$

## Existence and Uniqueness

Because of the bounded initial conditions, [1, Theorem 3.1] implies that the following condition on  $Y$  will imply existence and uniqueness:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0,$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta,$$

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha,$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x/\alpha.$$

## Some Results for a Globally Lipschitz Drift Term

We first prove some results assuming that the drift term,  $\sigma$  is globally Lipschitz:

$$|\sigma(x) - \sigma(y)| \leq L(\sigma)|x - y|, \quad x, y \in \mathbb{R}^d.$$

We also define

## Proposition

Suppose that  $d < 2\alpha$  and  $\rho = \rho(d) > 0$ . Then there exists a universal constant  $K := K(\alpha, \beta, \gamma, d) > 0$  such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

Then we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (0.3)$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any  $T > 0$ ,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[ 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (0.4)$$

Suppose that  $0 < \theta < (\Theta - d) \wedge 2$  and  $\rho = \rho(d) > 0$ . Let  $T > 0$  and  $t, r \in [0, T]$ . Then for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

and any  $x, z \in \mathbb{R}^d$  we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t - r|^q + |x - z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[ \mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left( 2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (0.5)$$

where,

1.  $0 < q \leq \rho$  under  $\beta \in (0, 1]$  and  $\gamma \in (0, 1 - \beta]$ ,
2.  $0 < q < \rho$  under  $\beta \in (0, 2)$  and  $\gamma > 0$ ,

and

$$\mathcal{M}_1 = \sqrt{\rho}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{\rho}L(\sigma).$$

## Holder Continuity

The solution,  $u$ , has a version, still denoted by  $u$ , that is  $\eta_1$ -Hölder continuous in time and  $\eta_2$ -Hölder continuous in space with  $0 < \eta_1 < q/2$  and  $0 < \eta_2 < \theta/2$  where  $0 < \theta < (\Theta - d) \wedge 2$  and

1.  $0 < q \leq \rho$  under  $\beta \in (0, 1]$  and  $\gamma \in (0, 1 - \beta]$ ,
2.  $0 < q < \rho$  under  $\beta \in (0, 2)$  and  $\gamma > 0$ ,

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

then we have that

$$\begin{aligned} & \mathbb{E} \left[ \sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \\ & \leq 2^{p-1} + C(p, \theta, T, R) \left[ \mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left( 2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right]. \end{aligned}$$

## Locally Lipschitz Diffusion Term

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[ \sigma(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases}$$

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta,$$

where  $\sigma_1 = \sigma(0)$  and  $\sigma_2 > 0$ .

### Theorem

Let  $M, T > 0$ . Under Dalang's condition,  $\rho(d) > 0$  and  $d < 2\alpha$ , there exists a random field solution for  $|x| \leq M$  denoted as  $(u(t, x) : (t, x) \in [0, T] \times [-M, M])$ . This solution is unique and satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, \text{ a.s.}$$

## Proof Sketch

We consider a truncated version of the equation where  $\sigma$  replaced by  $\sigma_N$ :

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}}.$$

We see that  $\sigma_N$  is Lipschitz:

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|,$$

so there exists a unique solution which is Holder continuous, which we denote as  $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$ . We define,

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T.$$

We see that on  $\{t < \tau_N\}$ ,  $u_N(t, x) = u_{N+k}(t, x)$  for any  $k \in \mathbb{N}$ .

## Proof Sketch

We can show that  $\sup_N \tau_N = T$  which implies that  $\{t < \tau_N\} \uparrow \Omega$ . On each  $\{t < \tau_N\}$ , define  $(u(t, x) : (t, x) \in [0, T) \times \mathbb{R})$  by  $u(t, x) = u_N(t, x)$  and hence  $u(t, x) = u_{N+k}(t, x)$  for any  $k \in \mathbb{N}$ . This implies that on  $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on  $\{t < \tau_N\}$ , we have that  $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$  and so  $u$  satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since  $\{t < \tau_N\} \uparrow \Omega$ , we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T) \times \mathbb{R} \quad (0.6)$$

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